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Adenylnucleotide-mediated binding of the PII-like protein SbtB contributes to controlling activity of the cyanobacterial bicarbonate transporter SbtA

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Abstract

Cyanobacteria have evolved a remarkably powerful CO₂ concentrating mechanism (CCM), enabling high photosynthetic rates in environments with limited inorganic carbon (Ci). Therefore, this CCM is a promising system for integration into higher plant chloroplasts to boost photosynthetic efficiency and yield. The CCM depends on active Ci uptake, facilitated by bicarbonate transporters and CO₂ pumps, to elevate CO₂ concentration around the active sites of the primary CO₂ fixing enzyme, Rubisco, which is encapsulated in cytoplasmic micro-compartments (carboxysomes). The essential CCM proteins have been identified, but the molecular signals and regulators that coordinate function in response to light, Ci availability and other environmental cues are largely unknown. Here, we provide evidence, based on a novel *in vitro* binding system, for a role of the PII-like SbtB protein in regulating Ci uptake by the bicarbonate transporter, SbtA, in response to the cellular adenylate energy charge (AEC) through dynamic protein-protein interaction. Binding of the SbtA and SbtB proteins from two phylogenetically distant species, *Cyanobium* sp. PCC7001 and *Synechococcus elongatus* PCC7942, was inhibited by high ATP, and promoted by low [ATP]:[ADP or AMP] ratios *in vitro*, consistent with a sensory response to the AEC mediated through adenylnucleotide ligand-specific conformation changes in SbtB. *In vivo*, cell cultures of *S. elongatus* showed up to 70% SbtB-dependent down-regulation of SbtA bicarbonate uptake activity specifically in the light activation phase during transitions from dark to low light when low cellular AEC is expected to limit metabolic activity. This suggests SbtB may function as a curfew protein during prolonged low cellular AEC and photosynthetically unfavourable conditions to prevent energetically futile and physiologically disadvantageous activation of SbtA.

eLife assessment

This study is of **fundamental** importance, addressing the regulation of the carbon concentrating mechanism in cyanobacteria. It is a well-controlled investigation of low affinity regulatory binding of small molecules, processes that are typically difficult to examine. The work provides **compelling** evidence that the adenylate pool, rather than any single metabolite, regulates a key bicarbonate transporter (SbtA) to provide efficient bicarbonate supply while preventing futile cycling that can result from escape of unfixed CO₂.

Introduction

The cyanobacterial CO₂ concentrating mechanism (CCM) is one of the most efficient CCMs known to date. It has been fundamental to their adaptability and successful competition in environments providing limited inorganic carbon (Ci) supply for oxygenic photosynthesis and is reflected in an estimated contribution of 30-80% global primary productivity in the oceans (Liu et al., 1997; Field et al., 1998). Therefore, recent efforts to enhance photosynthetic carbon fixation in crop species have focussed on introducing a synthetic cyanobacterial CCM into chloroplasts, which has been predicted to translate into 36-60% higher crop productivity (Price et al., 2013; McGrath and Long, 2014; Long et al., 2016; Meyer et al., 2016). The cyanobacterial CCM enables active uptake of two Ci species, CO₂ and bicarbonate (HCO₃⁻), which leads to accumulation of a cytoplasmic bicarbonate pool as large as 20-40 mM (Badger and Andrews, 1982; Price and Badger, 1989). This bicarbonate pool supports carbonic anhydrase (CA)-mediated elevation of the CO₂ concentration around the primary CO₂-fixing enzyme ribulose-1,5-bisphosphate carboxylase/oxygenase (Rubisco) inside proteinaceous micro-compartments known as carboxysomes. Consequently, Rubisco carboxylation is enhanced over oxygenation, suppressing energetically costly photorespiration (for review see (Badger and Price, 1992; Kaplan and Reinhold, 1999; Price et al., 2008; Espie and Kimber, 2011; Price, 2011)). To date, five Ci uptake systems have been verified, varying with respect to energization, substrate affinity, flux rates and induction by low Ci conditions (Price, 2011). Two light-dependent, thylakoid-located CO₂ pumps or energy-coupled CAs, Ndh-1₃ and Ndh-1₄ (Maeda et al., 2002; Ohkawa et al., 2002), convert external and internally recaptured CO₂ to HCO₃⁻, energized by NADPH and reduced ferredoxin (Ogawa et al., 1985; Maeda et al., 2002; Price et al., 2008). Three types of plasma membrane-located bicarbonate transporters, the ATP-dependent BCT1 complex (Omata et al., 1999) and the HCO₃⁻/Na⁺ symporters BicA (Price et al., 2004) and SbtA (Shibata et al., 2002), transfer bicarbonate directly from the external environment into the cytoplasm upon activation by low Ci conditions and light (Sültemeyer et al., 1998; McGinn et al., 2003). Of particular interest for a synthetic CCM is SbtA due to its ability to function in nonphotosynthetic *E. coli* (Du et al., 2014). SbtA requires a Na⁺ gradient directed from the outside to the inside of the cell for activity (Shibata et al., 2002; Price et al., 2004). Both CO₂ uptake and HCO₃⁻ transport are activated by light within 5 and 30 s, respectively, (Badger and Andrews, 1982; Price et al., 2011). Early on, Kaplan *et al.* reported a gradual inactivation of Ci uptake in the dark, suggesting a link to the state of photosynthetic electron transport or a redox signal (Kaplan et al., 1987), but the molecular mechanisms regulating the activity of specific bicarbonate transporters are largely unknown.

A highly conserved feature of cyanobacterial genomes is the presence of the *sbtB* gene, encoding the PII-like SbtB protein, in the immediate gene neighbourhood of the *sbtA* gene (Rae et al., 2011). Both genes are co-expressed under Ci limitation in *Synechococcus elongatus* PCC7942 and *Synechocystis* sp. PCC 6803 (Woodger et al., 2003; Schwarz et al., 2011), suggesting a functional connection between the SbtA and SbtB proteins. This has been supported by evidence from co-expression of several SbtA and SbtB proteins in *E. coli*, where SbtA-mediated bicarbonate uptake was specifically inactivated in the presence of the cognate SbtB protein, and a physical interaction was shown for the SbtA and SbtB proteins from *S. elongatus* PCC7942 (hereafter SbtA₇₉₄₂ and SbtB₇₉₄₂, respectively) (Du et al., 2014). Similarly, a study by Selim et al. (Selim et al., 2018) indicated SbtA and SbtB from *Synechocystis* sp. PCC6803 (hereafter SbtA₆₈₀₃ and SbtB₆₈₀₃, respectively) interacted in response to Ci availability, which was enhanced by addition of ADP or AMP and diminished by cAMP. It was suggested the cAMP:AMP ratio may indirectly signal intracellular Ci levels and thus regulate SbtA₆₈₀₃:SbtB₆₈₀₃ complex formation, but effects on SbtA activity were not directly assessed. Meanwhile, a broader role for SbtB₆₈₀₃ in Ci and metabolic sensing has emerged. The same study (Selim et al., 2018) showed that the *sbtB6803* knockout mutant was impaired in acclimation to high Ci conditions, remaining locked in the high Ci affinity state typical of low Ci acclimation. Moreover, SbtB₆₈₀₃ was implicated in far red light induced inhibition of photosynthetic oxygen evolution (Oren et al., 2021) and in bis-(3',5')-cyclic diadenosine monophosphate (c-di-AMP) and redox state sensing linked to control of glycogen synthesis and growth performance of cyanobacterial cells during day/night cycles (Selim et al., 2021; Selim et al., 2023).

Structural evidence for SbtA:SbtB interaction comes from X-ray crystallography and cryo-electron microscopy, which resolved trimeric SbtA₆₈₀₃ bound to SbtB₆₈₀₃ trimers in the presence of AMP as a ligand to SbtB (Fang et al., 2021; Liu et al., 2021). Furthermore, SbtB proteins have been classified as non-canonical PII-like members of the multi-functional PII and PII-like protein superfamily that consists of transcriptional and protein activity regulators, environmental signal transducers or sensors, widely-distributed across bacteria, archaea and plants (Radchenko et al., 2010). In cyanobacteria, there is substantial evidence for a central role of PII proteins in sensing and balancing the carbon and nitrogen status of the cell as part of regulatory networks that affect Ci and N acquisition (Forchhammer and Selim, 2020). In crystals, SbtB proteins form homotrimers typical of PII with clefts between monomers forming binding sites for small effector molecules (Forchhammer and Lüddecke, 2016) that may bind in a competitive manner or cooperatively. Common ligands for PII proteins are ATP, ADP, Mg²⁺ and 2-oxoglutarate (2OG) which impact on protein conformation, particularly the T-loop, and the interaction with target proteins (Radchenko et al., 2010; Huergo et al., 2013; Zeth et al., 2014; Forchhammer and Lüddecke, 2016; Forchhammer and Selim, 2020). Ligands for cyanobacterial PII-like proteins include ADP and bicarbonate for the carboxysome-associated CPII from *Thiomonas intermedia* (Wheatley et al., 2016), and adenylnucleotides for the SbtB₆₈₀₃ protein, i.e. ATP, ADP, AMP, cAMP and c-di-AMP (Selim et al., 2018; Selim et al., 2021) and the SbtB protein from *Cyanobium* sp. PCC7001 (SbtB₇₀₀₁), i.e. cAMP, AMP, ADP, ATP and Ca²⁺ ATP (Kaczmarek et al., 2019). SbtB₆₈₀₃ and SbtB₇₀₀₁ share the basic architecture, but differ in ligand-coordinating residues, ligand affinities and ligand-induced conformation changes. Recently, SbtB₆₈₀₃ was shown to exert apyrase activity, slowly converting ATP ligands to AMP in a redox state-dependent manner (Selim et al., 2023). Apyrase activity was tightly linked to the oxidation and disulfide bond formation between cysteines of the C-terminal R-loop, which is an extension that is present on a subset of SbtB proteins but not in SbtB₇₀₀₁ and SbtB₇₉₄₂. Therefore, one may expect fundamental mechanistic differences from SbtB₆₈₀₃.

The energy status of living cells is reflected in the relative abundance rather than absolute concentrations of the adenylnucleotides ATP, ADP and AMP and referred to as the cellular

adenylate energy charge (AEC), described quantitatively as $([ATP] + 0.5[ADP]) / ([ATP] + [ADP] + [AMP])$ (Chapman et al., 1971). Intriguingly, SbtB₇₀₀₁ had 18 to 38-fold higher binding affinities for ATP over ADP, AMP and cAMP in the presence of Ca²⁺ (Kaczmariski et al., 2019), which suggested to us certain SbtB proteins may sense the AEC directly and dynamically interact with SbtA in response to changes in adenylate nucleotide ratios. We propose that the adenylate nucleotide ligand-induced structural changes observed for SbtB proteins determine SbtA:SbtB complex formation and control of the SbtA activation state. Here, we demonstrate that the adenylate nucleotides ATP, ADP, AMP and cAMP regulate *in vitro* binding of SbtA (SbtA₇₀₀₁) from *Cyanobium* sp. PCC7001 and SbtA₇₉₄₂ to their cognate SbtB proteins, and present *in vivo* effects of SbtB on SbtA activity in cyanobacteria linked to light-induced changes in AEC, consistent with a role of SbtB as AEC sensor and response regulator.

Results

SbtA:SbtB complex formation is affected specifically by adenylate nucleotides

For SbtA:SbtB interaction studies, we co-expressed SbtA₇₀₀₁, or SbtA₇₉₄₂, with their cognate SbtB proteins which were C-terminally tagged with human influenza hemagglutinin (HA) epitope and 5 or 6 histidines (His) in the *E. coli* strain DH5α (*E. coli*) and in a bicarbonate transporter-deficient *Synechococcus elongatus* PCC7942 mutant (*Synechococcus* ΔCS). The SbtB₇₀₀₁ and SbtB₇₉₄₂ protein were identified immunologically at 13 kDa, consistent with their calculated molecular weights, but the apparent molecular weights of SbtA₇₀₀₁ and SbtA₇₉₄₂ monomers (23 kDa and 29 kDa, respectively) were 11 kDa lower than their calculated molecular weights, as is commonly observed for membrane proteins and attributed to altered protein-SDS detergent interactions (Rath et al., 2009). Specific binding of SbtA to SbtB proteins was evaluated *in vitro* by metal immobilized chromatography (IMAC) using membrane-enriched, native protein extracts that contained both SbtA and HAHis-tagged SbtB proteins (Fig. S1). Only the HAHis-SbtB but no SbtA protein was found to attach to Ni²⁺ coated beads which ensured co-elution of SbtA and SbtB from the beads was exclusively due to the interaction of the two proteins. The *in vitro* binding assays without addition of effectors showed that SbtA and SbtB were largely dissociated in native protein extracts (Figs. 1, S2), which led us to focus on the association of SbtA:SbtB complexes in response to a range of potential effector molecules and co-factors reported to interact with PII and PII-like proteins. Based on our previous observation that SbtB₇₀₀₁ binds ATP, ADP, AMP and cAMP with micromolar affinities (Kaczmariski et al., 2019), we expected SbtA:SbtB complex formation and dissociation be affected by these molecules. Since Ca²⁺ enhanced binding of ATP to SbtB₇₀₀₁ *in vitro* (Kaczmariski et al., 2019), we also included 200 nM Ca²⁺ in binding assays as a potential cofactor, which was equivalent to the basal Ca²⁺ levels in cyanobacterial cells (Leganés et al., 2009) did not facilitate SbtA:SbtB binding by itself (Fig. S2).

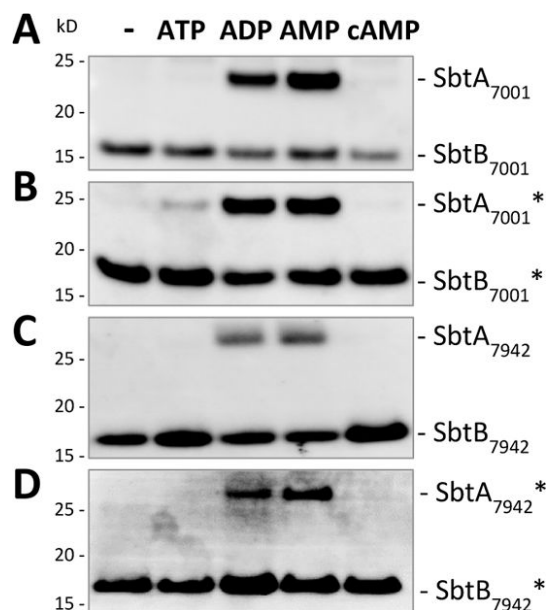


Fig. 1.

Association of SbtA and cognate SbtB proteins in response to adenylnucleotides.

The SbtA₇₀₀₁-SbtB₇₀₀₁ and SbtA₇₉₄₂-SbtB₇₉₄₂ pairs were expressed in *E. coli* (**A, C**) and *Synechococcus* Δ CS (denoted by *; **B, D**). The HAHis-tagged SbtB proteins were immobilized to examine interaction with SbtA proteins and adenylnucleotides (2 mM ATP, ADP, AMP, or cAMP) during *in vitro* IMAC binding assays. The HAHis-SbtB protein and corresponding amounts of SbtA retained in SbtA:SbtB complexes were detected by immunoblot analyses. Representative results are shown of n=3-8 experiments per adenylnucleotide treatment.

Under these conditions, the association of stable SbtA₇₀₀₁:SbtB₇₀₀₁ and SbtA₇₉₄₂:SbtB₇₉₄₂ complexes was observed exclusively in the presence of 2 mM ADP or AMP (**Fig. 1**). Conversely, SbtA and SbtB remained fully dissociated in the presence of 2 mM ATP or cAMP (**Fig. 1**).

In addition, a potential link between stabilization of SbtA:SbtB complexes and photosynthetic activity was tested, but SbtA:SbtB binding was not affected by the Calvin-Benson-Basham (CBB) cycle intermediates ribulose-1,5-bisphosphate (RUBP), 3-phosphoglycerate (3PGA) (**Fig. 2**). Similarly, 2OG which is generated in the tricarboxylic acid (TCA) cycle and acts a master regulatory metabolite coordinating carbon and nitrogen metabolism in association with canonical PII proteins (Huergo and Dixon, 2015), did not facilitate SbtA:SbtB complex formation (**Fig. 2**). Combined with ATP or ADP, 2OG had no synergistic effect (**Fig. 2**), in contrast to the prerequisite binding of Mg^{2+} ATP to the *E. coli* PII protein for 2OG to become effective (Radchenko et al., 2013). Furthermore, we investigated possible substrate-level regulation of SbtA activity via HCO_3^- -dependent complex formation with SbtB. HCO_3^- had no effect on SbtA:SbtB binding irrespective of combinations with Ca^{2+} or Mg^{2+} , ATP or AMP (**Fig. S2, S3**). In summary, we observed that small molecule effectors unable to bind to SbtB₇₀₀₁ (Kaczmarek et al., 2019), were equally ineffective in mediating the binding of either SbtA-SbtB pair. It is notable that SbtA-SbtB-effector interactions were very similar for proteins expressed in *E. coli* compared to *Synechococcus* Δ CS (**Fig. 1, 2**), which suggests proteins were not subject to species-dependent, post-translational modifications and validates use of *E. coli* as heterologous expression system.

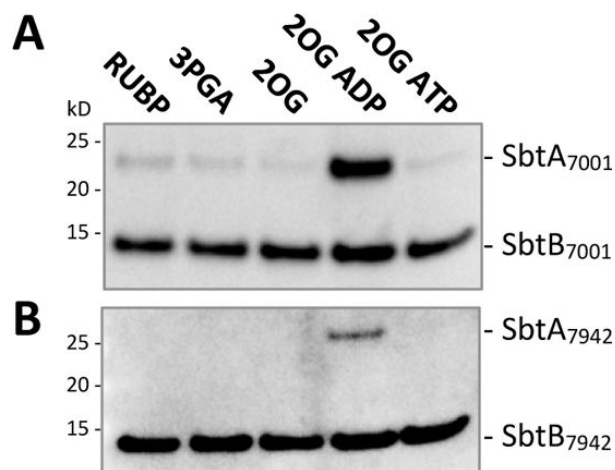


Fig. 2.

The association of SbtA and cognate SbtB proteins is not directly linked to key metabolites of the CBB cycle (RUBP, 3PGA) and the TCA cycle (2OG).

In vitro IMAC binding assays were performed with the SbtA₇₀₀₁-SbtB₇₀₀₁ (A) and SbtA₇₉₄₂-SbtB₇₉₄₂ (B) pairs expressed in *E. coli* in the presence of 2 mM RUBP, 3PGA or 2OG, and 2 mM 2OG combined with 2 mM ADP, or ATP (2OG ADP and 2OG ATP, respectively). The representative immunoblot images show HAHis-tagged SbtB proteins and the SbtA proteins retained in SbtA:SbtB complexes; n=2.

The AEC simulated by adenyl nucleotide ratios determined the SbtA:SbtB complex formation

The fact that SbtA:SbtB complexes were destabilized by ATP and stabilized by either ADP or AMP, strongly suggested complex formation would be impacted by changes in the cellular adenylate energy charge (AEC). Therefore, we simulated defined energy charge environments *in vitro* with different molar ratios of ATP:ADP or ATP:AMP within a constant total adenyl nucleotide pool of 2 mM. Both SbtA-SbtB pairs, irrespective of expression in *E. coli* or *Synechococcus* ΔCS, showed decreasing amounts of SbtA bound to SbtB with an increasingly higher proportion of ATP over ADP (Fig. 3A-D) and AMP (Fig. 3E,F). Since both SbtA-SbtB pairs were unbound in the presence of cAMP (Fig. 1), we replaced ATP with cAMP to test whether SbtA:SbtB binding could be affected by the molar ratios of cAMP to ADP or AMP, similar to what has been reported for the SbtA₆₈₀₃-SbtB₆₈₀₃ pair (Selim et al., 2018). It was indeed possible to impair SbtA:SbtB binding with high cAMP:ADP or AMP ratios (Fig. 4). However, to evaluate the effects of cAMP in relation to ATP, it is important to recognize that ATP concentrations up to 2 mM corresponded to the physiological range of intracellular ATP pools measured in several bacteria (Thauer et al., 1977; Yaginuma et al., 2014), whereas estimates of cAMP levels in cyanobacterial cells have been up to 1000-fold lower than the intracellular ATP levels, i.e. in the nano- to micromolar range (Ohmori and Okamoto, 2004). The *in vitro* cAMP levels reducing the amount of bound SbtA to 50% was 1.16 mM (Fig. 5B, AR₅₀ of SbtA₇₉₄₂ cAMP:AMP) which is likely to exceed the physiological range of intracellular concentrations by an order of magnitude. The lowest tested cAMP to ADP or AMP ratio of 0.2 (95 μM cAMP: 1.905 mM ADP or AMP) is within physiologically relevant range and cAMP did not affect the association of either SbtA-SbtB pair. Therefore, changes in cAMP concentrations expected to occur *in vivo* are likely inconsequential compared to the variations in intracellular ATP levels and the AEC.

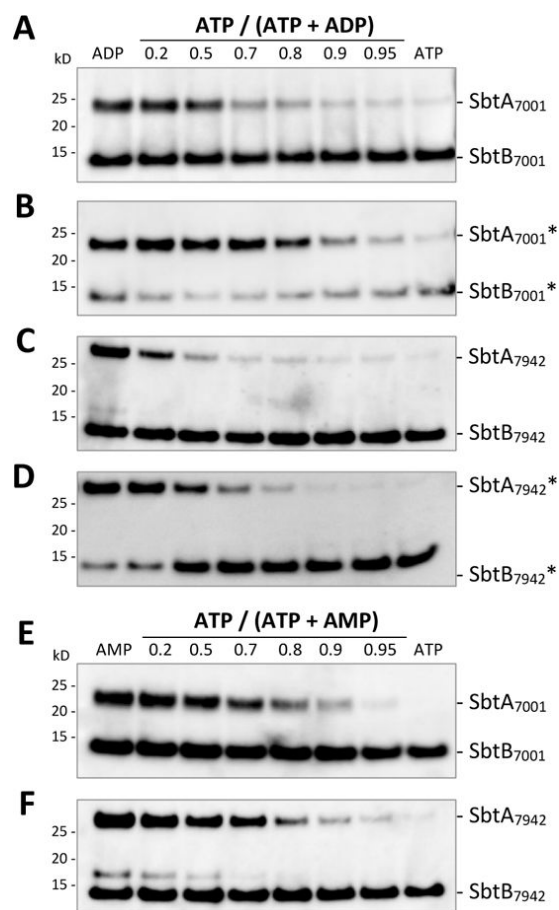


Fig. 3.

Effects of ATP:ADP and ATP:AMP ratios on the association of SbtA and SbtB proteins.

The SbtA₇₀₀₁-SbtB₇₀₀₁ (**A, C, E**) and SbtA₇₉₄₂-SbtB₇₉₄₂ (**B, D, F**) pairs were expressed in *E. coli* (**A, C, E, F**) and in *Synechococcus* Δ CS (denoted by *; **B, D**). Variation in energy charge was simulated during *in vitro* IMAC binding assays by different molar ratios of ATP:ADP (**A-D**) and ATP:AMP (**E, F**), with a total adenyl nucleotide concentration of 2 mM. Representative immunoblot images of HAHis-tagged SbtB and SbtA proteins retained in SbtA:SbtB complexes of n=3-4 per adenyl nucleotide treatment. The 17 kD-protein above SbtB₇₉₄₂ is a fragment occasionally detected with the anti-SbtA antibody (**C, F**).

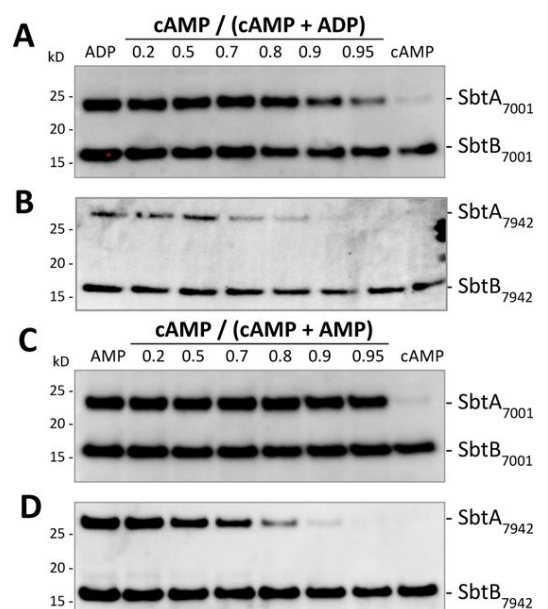


Fig. 4.

Effects of the cAMP:ADP and cAMP:AMP ratios on the association of SbtA and SbtB proteins.

In vitro IMAC binding assays were performed across a range of cAMP:ADP (**A, B**) and cAMP:AMP (**C, D**) ratios, with a total adenyl nucleotide concentration of 2 mM. The SbtA₇₀₀₁-SbtB₇₀₀₁ (**A, C**) and SbtA₇₉₄₂-SbtB₇₉₄₂ (**B, D**) pairs were expressed in *E. coli*. The immunoblot images show HAHis-tagged SbtB and SbtA proteins retained in SbtA:SbtB complexes and are representative of three independent experiments per adenyl nucleotide treatment.

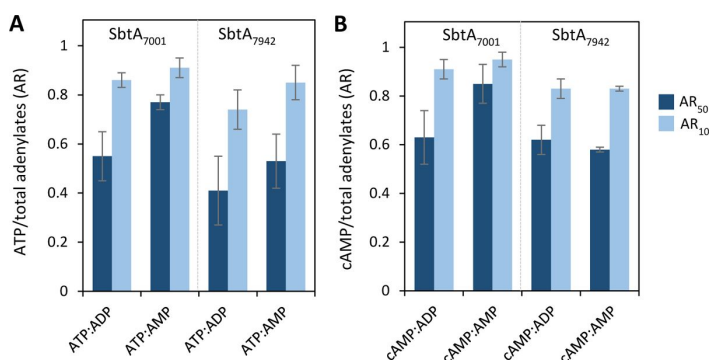


Fig. 5.

Quantitative estimates for the association of SbtA and SbtB proteins in response to different adenylnucleotide ratios (AR).

The ratios of ATP:ADP or AMP (**A**) and cAMP:ADP or AMP (**B**) that supported 50% (AR₅₀) and 10% (AR₁₀) of SbtA bound to SbtB were estimated from sigmoidal logistic curve fits (Fig. S4). AR values are means ($\pm$ SE) of 3-4 independent experiments. AR values were not significantly different across each AR category at a $P=0.05$ threshold (AR₅₀ $P=0.232$, AR₁₀ $P=0.275$).

To compare the association of SbtA with SbtB between adenylnucleotide treatments quantitatively, we estimated the relative changes in SbtA abundance from densitometric analyses of Western blot images (pixel volumes) against ATP:ADP or AMP and cAMP:ADP or AMP ratios (Fig. S4). The association of SbtA with SbtB was non-linearly correlated with the adenylnucleotide ratio and fitted best by a sigmoidal logistic model, typical of cooperative ligand binding (mean adjusted R^2 values of 0.93 ± 0.02 ; Fig. S4). Comparing the adenylnucleotide ratios supporting 50% (AR₅₀) or 10% (AR₁₀) of the maximum levels of SbtA bound to SbtB, derived from curve fits in Fig. S4, differences between the two SbtA-SbtB pairs and in response to the four different adenylnucleotide combinations tested were statistically not significant (Fig. 5). However, even though this implied very similar binding affinities of SbtB₇₀₀₁ and SbtB₇₉₄₂ for those adenylnucleotides as well as comparable ligand-induced binding to the cognitive SbtA proteins, SbtA₇₉₄₂ seemed to be dissociated at slightly lower ATP:ADP, cAMP:ADP and cAMP: AMP ratios than SbtA₇₀₀₁.

SbtA activity *in vivo* is modulated by SbtB and the light-stimulated adenylate energy charge

Inside photosynthetic cyanobacterial cells, SbtA and SbtB proteins are both present when Ci supply is low (Fig. S1A). In nonphotosynthetic *E. coli*, SbtA appeared to be constitutively inactivated in the presence of SbtB (Du et al., 2014), which implied AEC levels were low under those experimental conditions. In cyanobacteria, however, SbtA is active irrespective of SbtB when photosynthesis is fully light-activated and cells operate at high AEC levels (Figs. 6, S6, S7). Therefore, it was of interest to investigate SbtA function in low Ci combined with low AEC states which are prominent during dark-to-low light transitions.

Photosynthetic performance and uptake of ^{14}C -labelled HCO_3^- in *Synechococcus* ΔCS expressing SbtA₇₉₄₂ as the only HCO_3^- uptake system was compared either with or without SbtB₇₉₄₂ (SbtAB and SbtA, respectively). Note that concurrent CO_2 uptake by Ndh-1_{3/4} complexes was eliminated by addition of the carbonic anhydrase inhibitor ethoxzolamide

(EZ), which does not affect SbtA function (Fig. S5), and measurements were performed at pH 8.0 where HCO_3^- represents 98% of the Ci pool. Under these conditions, the cells fully depended on SbtA activity for inorganic carbon supply. At light intensities above the light compensation point, O_2 evolution rates directly reflected the activity of SbtA₇₉₄₂. In Ci-starved cell cultures steady-state photosynthetic rates in response to incremental additions of bicarbonate were essentially the same for SbtA and SbtAB with substrate affinities ($K_{0.5\text{Ci}}$; 9 ± 4 and 30 ± 5 μM Ci, respectively) and maximum rates of photosynthetic O_2 evolution (V_{maxO_2} ; 164 ± 19 and 167 ± 12 $\mu\text{mol mg Chl}a^{-1} \text{h}^{-1}$, respectively) (Fig. S6). Similarly, alternating light intensities between 1400 and 200 $\mu\text{mol photons m}^{-2} \text{s}^{-1}$ in the presence of saturating Ci levels (250–500 μM) did not lead to discernible differences between SbtA and SbtAB in steady state HCO_3^- uptake or photosynthetic O_2 evolution rates (Fig. S7). Together, this suggested SbtB is not required for and does not regulate SbtA₇₉₄₂ activity once photosynthesis has been fully activated in response to light-intensity driven changes in carboxylation rates and Ci demand.

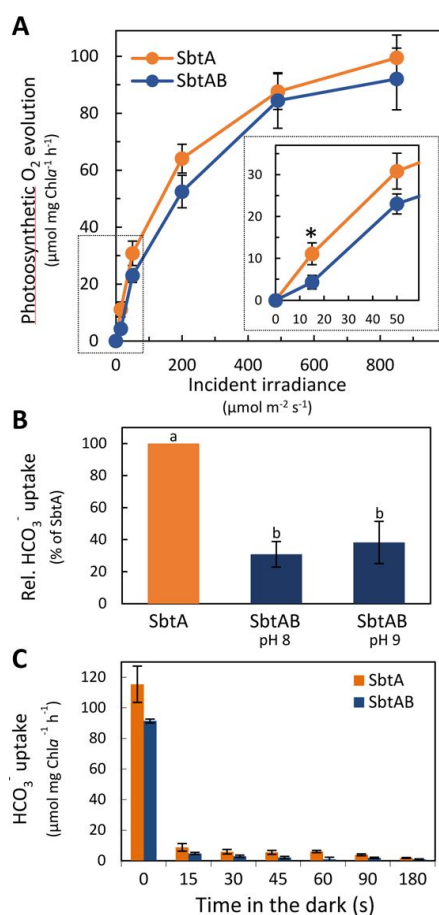


Fig. 6

Light-dependent bicarbonate uptake by SbtA is modulated by SbtB in *Synechococcus* ACS.

The SbtA₇₉₄₂ protein was expressed alone (SbtA) or with SbtB₇₉₄₂ (SbtAB) in *Synechococcus* ACS (ΔCS) at low Ci in the light. CO_2 uptake was inhibited by EZ to restrict Ci acquisition to bicarbonate uptake by SbtA. (A) HCO_3^- uptake-dependent gross photosynthetic O_2 evolution rates in response to stepwise increased light intensity after 1 h dark-adaptation. Respiratory O_2 consumption was the same for SbtA and SbtAB (27 ± 4 and 27 ± 3 , respectively). Values are means $\pm$ SE; $n=4-5$. (B) Induction of HCO_3^- transport activity of SbtA in dark-adapted cells immediately after dark-to-low light ($30 \mu\text{mol m}^{-2} \text{s}^{-1}$) transitions at pH 8 and pH 9 during 30–60 s pulses of ^{14}C HCO_3^- uptake, using silicon oil centrifugation-filtration. HCO_3^- uptake rates are shown relative to cells expressing SbtA alone; $n=4-5$. Relative HCO_3^- uptake rates were calculated by subtraction of the unspecific incorporation of ^{14}C -label, mainly due to extracellular ^{14}C HCO_3^- in the water shell surrounding each cell and a very small amount of $^{14}\text{CO}_2$ diffusion into the cells. This was derived from ^{14}C -label incorporation into Ci uptake inhibited ΔCS (38 ± 6 (pH 8) and 42 ± 8 (pH 9)). (C) HCO_3^- uptake rates normalized to Chl a content at various time points after a light ($400 \mu\text{mol m}^{-2} \text{s}^{-1}$) to dark shift. Values are means $\pm$ SE; $n=3-5$. Statistically significant differences ($P < 0.05$) are denoted by an asterisk (A) or different letters (B).

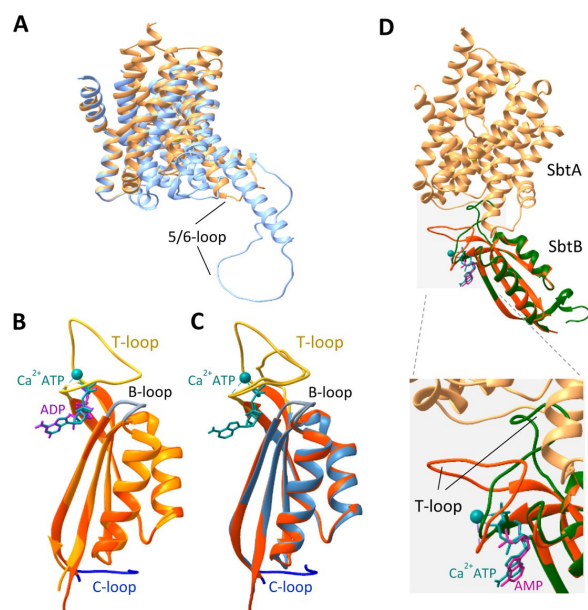


Fig 7.

Structural comparison of the overall fold of SbtA and SbtB protein models.

(A) Superposition of homology models of SbtA₇₀₀₁ (light orange) and SbtA₇₉₄₂ (light blue). The core domains and arrangement of the transmembrane helices are well matched but the 5/6 loop connecting transmembrane helix 5 and 6 is divergent, protruding further into the cytoplasm on SbtA₇₉₄₂. (B) Superposition of the three-dimensional models based on crystal structures of SbtB₇₀₀₁ with Ca²⁺ and ATP ligands (red-orange, SbtB₇₀₀₁-CaATP; PDB: 6MM2) and SbtB₇₀₀₁ with ADP bound (orange, SbtB₇₀₀₁-ADP; PDB: 6MMC), demonstrating that ligand-induced structural differences affect primarily the T-loop. The T-loop assumes a closed conformation in SbtB₇₀₀₁-CaATP and an open, disordered state in SbtB₇₀₀₁-ADP which was not resolved in crystals as indicated by the dashed line. (C) Superposition the of the homology model of the apo-SbtB₇₉₄₂ monomer (blue) on SbtB₇₀₀₁-CaATP; both SbtB proteins are predicted to be nearly identical in three-dimensional structures. (D) Superposition of the SbtA₇₀₀₁ and SbtB₇₀₀₁-CaATP onto a SbtA₆₈₀₃ monomer (PDB: 7EGK, chain C) associated with a SbtB₆₈₀-AMP monomer (chain D) which were extracted from the trimeric SbtAB₆₈₀₃ cryo-EM structure (PDB: 7EGK). Since the three-dimensional architecture of SbtA₇₀₀₁ and SbtA₆₈₀₃ are highly similar (Fig. S12), only SbtA₇₀₀₁ is shown for clarity. The core structure of SbtB₇₀₀₁-CaATP and SbtB₆₈₀₃-AMP is nearly identical, but the T-loop conformations diverge in a ligand-specific manner consistent with the observed SbtA:SbtB interaction in this study (enlarged). Binding of AMP to SbtB₆₈₀₃ (green) promotes an open T-loop conformation which extends into SbtA consistent with SbtA:SbtB association, whereas Ca₂⁺ATP bound to SbtB₇₀₀₁ (red-orange) promotes folding of the T-loop away from SbtA.

In contrast, SbtB clearly modulated SbtA function in the initial phase of dark-to-low light transitions. After an hour of dark-adaptation, cells co-expressing SbtA and SbtB were unable to activate SbtA₇₉₄₂-dependent photosynthetic O₂ evolution to the same extent as cells expressing SbtA alone during the initially low light phase of light response measurements, when light levels were below or near the light compensation point of 10-15 $\mu\text{mol photons m}^{-2} \text{s}^{-1}$ (Fig. 6A). Low light activation of ¹⁴C-labelled HCO⁻ uptake by SbtA₇₉₄₂ clearly showed SbtA activation was strongly suppressed in the presence of SbtB₇₉₄₂ during the first 30-60 s after long-term dark-adaptation, reducing net SbtA activity to 31% and 38% of the activity observed in the absence of SbtB at pH 8 and pH 9, respectively (Fig. 6B). Since many photosynthetic processes are almost instantaneously inactivated in the dark, we tested whether SbtB affected [¹⁴C]HCO⁻ uptake by SbtA during light to dark transition. Interestingly, SbtA₇₉₄₂ was fully inactivated within 15 s after cells were shifted from light into the dark without any effect on the inactivation kinetics by the presence of the SbtB₇₉₄₂ protein (Fig. 6C). Altogether this suggests, SbtA activity is regulated at multiple levels, which may be SbtB-independent, but also involve SbtB and AEC coupled regulation of SbtA activity demonstrated by dark-to-light activation kinetics of SbtA.

In *Synechocystis sp.* PCC6803, deletion of SbtB₆₈₀₃ was shown to reduce growth under low Ci condition in low light (Selim et al., 2018), and a broader regulatory function of CCM components by SbtB was proposed. If this was the case in *Synechococcus* ΔCS , one would expect a negative effect on growth in cells expressing SbtA alone compared to those co-expressing SbtB and SbtA. However, photoautotrophic growth of cells expressing either

SbtA₇₉₄₂ or both SbtA₇₉₄₂ and SbtB₇₉₄₂, was very similar when supplied with air levels of CO₂ under diurnal light-dark cycles (Fig. S8), suggesting deletion of SbtB₇₉₄₂, unlike SbtB₆₈₀₃, did not impair photoautotrophic growth of *Synechococcus* ΔCS under the conditions that provide a relatively stable light environment and sufficient CO₂ and nutrient supply.

Discussion

Allosteric regulation of cyanobacterial bicarbonate transporters has been an unresolved question for quite some time with patchy evidence for diverse mechanisms and co-factors that may be involved. It has already been established that SbtB is capable of exerting negative regulation on SbtA bicarbonate uptake activity in a heterologous expression system (Du et al., 2014). Here, we provide evidence for regulation of two SbtA bicarbonate transporters by protein-protein interaction with their cognate PII-like SbtB proteins in response to the *in vitro* simulation and *in vivo* modulation of the adenylate energy charge. The SbtA-SbtB pairs originated from the transitional α-cyanobacterium *Cyanobium* sp. PCC7001 and the β-cyanobacterium *Synechococcus elongatus* PCC7942, which are phylogenetically relatively distant species (Rae et al., 2011; Sánchez-Baracaldo et al., 2019) but share overall sequence homology at the amino acid level, with 60% identity between the two SbtA (Fig. S9A, S10) and 58% identity between the two SbtB (Fig. S9B, S11) proteins. Importantly, the modelled structures of SbtA₇₀₀₁ compared to SbtA₇₉₄₂ and SbtB₇₀₀₁ compared to SbtB₇₉₄₂ are nearly identical based on three-dimensional models (Fig. 7). Together with convincingly similar interaction profiles for both SbtA-SbtB pairs, we concluded they likely share the same overall regulatory mechanisms. It should be noted that about 50% of the cyanobacterial SbtB family, which includes the SbtB proteins studied here, lack a putative redox regulatory domain on the C-terminus, as typically found on SbtB of *Synechocystis* sp. PCC6803 (Selim et al., 2023). This makes the present two SbtAB pairs a simpler system for studying the nature of the primary signal(s) that direct allosteric inactivation/activation of SbtA.

The AEC rather than cAMP is the probable primary effector determining SbtA:SbtB interactions

Intracellular adenylnucleotide pools invariably contain ATP, ADP, AMP and cAMP, albeit relative concentrations may vary rapidly dependent on metabolic activity in response to environmental and developmental factors (De la Fuente et al., 2014). In cyanobacteria, cAMP is predominantly a secondary messenger participating in different signalling events, whereas the relative abundances of ATP, ADP and AMP define and communicate the cellular energy status. Notwithstanding, both processes could translate Ci availability and Ci demand into SbtB-mediated regulation of SbtA. In addition, relative abundances of cAMP, ATP and AMP in the adenylate pool are interdependently influenced by the activities of adenylate cyclase and phosphodiesterases which catalyze cAMP synthesis from ATP and hydrolytic degradation to AMP, respectively; depending on the species, adenylate cyclase activity is stimulated by N and P deficiency, osmotic stress, light quality and in some cases also bicarbonate or CO₂ (Cann, 2004; Hammer et al., 2006; Xu and Su, 2009; Agostoni and Montgomery, 2014), potentially linking cAMP and Ci sensing.

Initially, evidence from *Synechocystis* sp. PCC6803 suggested the cAMP:AMP ratios determined formation of SbtA:SbtB complexes and SbtA activity (Selim et al., 2018). SbtB₆₈₀₃ was proposed to be a novel cAMP receptor sensing the Ci status and mediating acclimation of the CCM to high Ci levels, based on impaired acclimation in mutant cell lines lacking SbtB.

In that study, high Ci acclimated cells showed the expected general downregulation of Ci uptake alongside relatively higher cAMP levels and cAMP:AMP ratios than in low Ci acclimated cells. However, SbtB₆₈₀₃ was dissociated from SbtA₆₈₀₃ which implied SbtA was inactive in its unbound state. Conversely, bicarbonate uptake was upregulated in low Ci acclimated cells with high AMP levels and low cAMP:AMP ratios, which favoured formation of SbtA:SbtB complexes, and therefore suggested SbtA was active when bound to SbtB. Those findings were difficult to reconcile with the simple interaction model for SbtB-mediated SbtA regulation in *E. coli*, where SbtA alone showed activity and co-expression of SbtA and SbtB inhibited SbtA-dependent bicarbonate uptake, including the SbtA₆₈₀₃-SbtB₆₈₀₃ pair (Du et al., 2014). Meanwhile, regulation of SbtA₆₈₀₃ has been revised based on the finding that SbtB₆₈₀₃ displays high affinity binding to SbtA only with AMP as ligand and when the R-loop Cys residues are oxidized, which appeared to be the only condition for T-loop conformation that permits binding to SbtA (Selim et al., 2023).

The binding analyses performed here (Fig. 3), and the fact that the R-loop is absent from SbtB₇₀₀₁ and SbtB₇₉₄₂ (Fig. S9B), clearly support the view that low levels of ATP and low ATP:ADP and AMP ratios, representing a low AEC, determine the association of SbtA₇₀₀₁-SbtB₇₀₀₁ and SbtA₇₉₄₂-SbtB₇₉₄₂ pairs rather than cAMP levels and redox state of SbtB. Considering, that the cAMP concentrations that effectively prevented the *in vitro* association of SbtA and SbtB were up to 1000-fold higher than physiological levels of cAMP in cyanobacteria (Ohmori and Okamoto, 2004), it seems unlikely the intracellular concentration of cAMP required for effective competition with ATP for binding sites on SbtB would occur *in vivo*. Furthermore, the 6-12 times higher binding affinity of SbtB₇₀₀₁ for ATP over cAMP, ADP or AMP (Kaczmarek et al., 2019) is consistent with the proposed AEC-dependent regulation of these SbtA-SbtB pairs. This AEC control of SbtA via binding/unbinding of SbtB might be a feature across many cyanobacteria, but further experimentation is required to confirm this.

The SbtA and SbtB proteins interact dynamically in response to the adenylate energy charge

ATP, ADP and AMP form the core of the energy system of living cells. The majority of biochemically available energy is linearly correlated with the molar fraction of ATP and ½ molar fraction of ADP of the total adenylate pool, but was experimentally simplified here by testing effects of ATP:ADP and ATP:AMP ratio effects separately (Fig. 3). Adenylate pools measured from a diverse range of tissues, organisms and environmental conditions suggest active growth requires an adenylate energy charge between 0.8 and 0.9, while a drop of the AEC below 0.5 is unsustainable (Chapman et al., 1971). Under favourable environmental conditions, the AEC is kept remarkably steady near 0.8 in cyanobacteria the light (Rust et al., 2011). Regulatory proteins often sense the AEC through competitive binding of ATP or ADP (Atkinson and Walton, 1967), which includes PII proteins that regulate target enzyme activity by undergoing specific conformational changes in response to binding ATP or ADP, which alters the interaction between the PII protein and its target and modulates the activity of the target protein (Radchenko et al., 2013; Zeth et al., 2014; Selim et al., 2019). This principle also appears to apply to the interaction of the PII-like SbtB proteins with respective SbtA transporters from both cyanobacterial species investigated. More than 90% of SbtA was unbound (proposed active state) at a simulated AEC of 0.74 to 0.91 (Figs. 3, 5) approximating measured ATP/(ATP+ADP) ratios of 0.8 to 0.9 reported for *Synechococcus elongatus* (formerly *Anacystis nidulans*) (Rust et al., 2011), recalculated from (Bornefeld and Simonis, 1974) and the thermophile *Synechococcus* strain Y-7c-s in medium light (250-270 $\mu\text{mol m}^{-2} \text{s}^{-1}$) (Kallas and Castenholz, 1982). This suggests, in the light, photosynthetically active cyanobacterial cells would maintain an almost fully active pool of SbtA, if the AEC was the defining parameter. The fact that SbtA₇₉₄₂ activity was the same for photosynthetically fully activated *Synechococcus* ΔCS strains expressing either SbtA or SbtA and SbtB (Fig. S6, S7) supports the

notion that SbtB had in fact fully dissociated from SbtA and did not limit SbtA activity at high AEC. Without photosynthetic ATP production in the dark, or during excessive cellular energy demand such as sudden decreases in light intensity and nutrient, pH or temperature stress, the AEC is likely to decrease substantially in cyanobacterial cells. In *S. elongatus*, the ATP:ADP ratio, was reduced to 0.55-0.6 about 10 min after light to dark transfer (Kallas and Castenholz, 1982; Rust et al., 2011). Based on our *in vitro* binding analysis, this suggests, on average more than 50% of both SbtA7001 and SbtA7942 should be bound to their respective SbtB proteins and form a stably-inactivated SbtA pool in prolonged darkness. During light exposure, after long dark periods, it can take from seconds to minutes for the cellular AEC to reach 0.8-0.9, dependent on light intensity and activation of CO₂ fixation by Rubisco and the CBB cycle enzymes. Indeed, low light (70-100 $\mu\text{mol m}^{-2} \text{s}^{-1}$) did not increase the AEC beyond 0.6 in thermophilic *Synechococcus* (Kallas and Castenholz, 1982).

Importantly, when *Synechococcus* ΔCS strains expressing either SbtA or SbtA and SbtB were analysed in dark-to-low light transitions, equivalent to raising AEC ratios from about 0.5 to 0.8-0.9, we detected a transient and marked retardation in SbtA-dependent bicarbonate uptake, decreasing SbtA activity by 60-70% when SbtB was present compared to SbtA alone in the initial low light (30-50 $\mu\text{mol m}^{-2} \text{s}^{-1}$) phase (Fig. 6A, B). Assuming an AEC of 0.6 in low light, similar to Kallas *et al.*, around 50% of SbtA should still have been associated with SbtB (Figs. 3, 5). In our view, this is a convincing indication that a rise in AEC is needed to displace the inhibitory SbtB curfew protein for SbtA activation during this transition and that this takes a few seconds to achieve. Despite the technical difficulty, future work looking at correlations between time-resolved AEC changes and bicarbonate uptake in *Synechococcus* ΔCS strains should be able to test this further.

Adenylnucleotide-dependent conformation changes of SbtB aligns with regulatory binding to SbtA

Structural information of SbtA:SbtB complexes and binding of adenylate ligands to SbtB strongly support the aforementioned AEC-dependent regulation of the SbtA:SbtB interaction *in vitro*. Inevitably, uncertainty exists in extrapolating from *in vitro* findings to implications for *in vivo* regulation. However, the case of regulation of SbtA by adenylate charge is at least feasible, especially for the significant class of SbtA-SbtB-containing species that lack the putative C-terminal redox regulatory domain on SbtB. The core structures and key amino acids involved in ligand binding that are thought to facilitate SbtA:SbtB interactions are highly conserved between crystal structures of SbtB₆₈₀₃ and SbtB₇₀₀₁ and homology models of SbtB₇₉₄₂ (Figs. 7B,C, S10) as well as crystallized SbtA₆₈₀₃ and homology models of SbtA₇₀₀₁ and SbtA₇₉₄₂ (Figs. 7A,C, S10A, S12), which lends some validity to propose a generalized model for the regulatory interaction of subsets of those SbtB proteins missing the R-loop and their cognate SbtA proteins. In PII proteins, the surface-exposed, flexible T-loop has been shown to mediate the regulatory interaction with target protein(s). Upon binding small effector molecules, ligand-induced conformation changes determine whether PII binds to its target. For example, binding of ATP to the PII protein from *Synechococcus elongatus* PCC7942 induced T-loop conformation changes that facilitated association with N-acetyl-glutamate kinase (NAGK) and arginine synthesis (Zeth et al., 2014; Selim et al., 2019), and the ADP-stabilized T-loop of the PII protein GlnK from *E. coli* enable formation of an inactivating complex with the ammonium transporter AMTB (Radchenko et al., 2010). Crystallographic data support a direct correlation, analogous to PII proteins, between adenylate-ligand induced changes in T-loop conformation of SbtB proteins (Selim et al., 2018; Kaczmarek et al., 2019; Fang et al., 2021; Selim et al., 2023) and can explain the opposite effects of ATP and ADP or AMP on SbtA:SbtB binding described here. The probability for each adenylate ligand to occupy a nucleotide binding site would depend on the binding affinity and the local concentration of the ligand. At high ATP:ADP or AMP ratios, when

predominantly Ca^{2+} ATP is likely to occupy the nucleotide binding sites of the SbtB₇₀₀₁ trimers, the T-loops are probably stabilized (Fig. 7B,C) through interactions of specific residues with the γ -phosphate of ATP, in a position close to the core overlapping the binding site (Kaczmarowski et al., 2019). When superimposed onto SbtB₆₈₀₃ in the trimeric SbtA-SbtB cryo-EM structure, it becomes apparent that the Ca^{2+} ATP-induced stable T-loop conformation of SbtB₇₀₀₁ interferes with binding to SbtA (Fig. 7D), which would allow SbtA to be fully active. In contrast, at low ATP:ADP or AMP ratios, occupation of the binding site mostly by ADP or AMP (Fig. 7B,C) would leave the T-loops in a disordered or extended conformation (Kaczmarowski et al., 2019; Fang et al., 2021), which would enable formation of inactivated SbtA:SbtB complexes.

A proposed model implicating SbtB in the regulation of SbtA activity in cyanobacteria

For cyanobacteria, it is a selective advantage to maximize photosynthetic CO_2 fixation and carbon gain for growth in the light. In the dark, photosynthetic CO_2 fixation ceases making Ci acquisition redundant and energetically futile. Additionally, continued accumulation of large amounts of bicarbonate may disrupt intracellular pH and ion homeostasis, particularly if $[\text{Na}^+]$ were to build up by $\text{HCO}_3^-/\text{Na}^+$ symport due to SbtA activity. Therefore, the co-ordinated regulation of Ci acquisition and photosynthetic activity may be an imperative for optimal survival and success of cyanobacteria. A hypothetical scenario for the regulation of bicarbonate uptake by SbtA in cyanobacteria is depicted in Fig. 8, assuming functional units of both SbtA and SbtB proteins are trimers as indicated by cryo-EM and crystal structures (Kaczmarowski et al., 2019; Fang et al., 2021). In *Synechococcus*, SbtA function is subject to several regulatory mechanisms operating at different time scales. In constant light and stress-free environments, bicarbonate uptake rates of SbtA are clearly a product of light intensity and Ci availability, independent of the SbtB protein (Figs. 6, S7). In light/dark cycles, SbtA is turned on and off within a few seconds (Fig. 6C) in sync with the light-induced changes in Ci demand for photosynthetic carboxylation, likely preventing the aforementioned complications associated with bicarbonate and Na^+ over-accumulation. The nature of the signals and signal transduction pathways triggering the fast activation-deactivation kinetics of SbtA may be complex and cannot be precisely known at this stage, but some details can be envisaged. SbtA-mediated bicarbonate uptake is immediately inactivated after a light-dark transition supposedly associated with a drop in the AEC, assuming a similar decrease of the ATP/(ATP+ADP) ratio from 0.85 to about 0.4 observed in *S. elongatus* previously (Bornefeld and Simonis, 1974; Takano et al., 2015). If AEC-induced binding of SbtB was the primary inactivation mechanism during the light-to-dark transition, one would expect de-regulation of SbtA activity in the absence of SbtB. Yet, SbtA-mediated bicarbonate uptake ceased within 15 s in the dark irrespective of the presence of SbtB (Fig. 6C), which suggested the initial fast inactivation of SbtA may be mediated by any one or a combination of other signals derived from the many dark-induced changes in the cell which may involve the more oxidized redox state (Tamoi et al., 2005), changes in net availability of ATP, in plasma membrane energization potentially involving Ca^{2+} signalling (Torrecilla et al., 2004), in the transmembrane Na^+ gradient, and in protein post-translational modifications such as phosphorylation (Spät et al., 2015; Angeleri et al., 2016). Many of these changes are likely faster than a possible low-AEC-induced SbtA:SbtB association. On the other hand, we did detect significant differences in the activation of bicarbonate uptake upon transition from dark to light (Fig. 6), indicating low AEC-induced formation of inactive SbtA:SbtB complexes in the dark delayed activation of SbtA until light-dependent increase in AEC was sufficient to facilitate the dissociation of SbtB and SbtA.

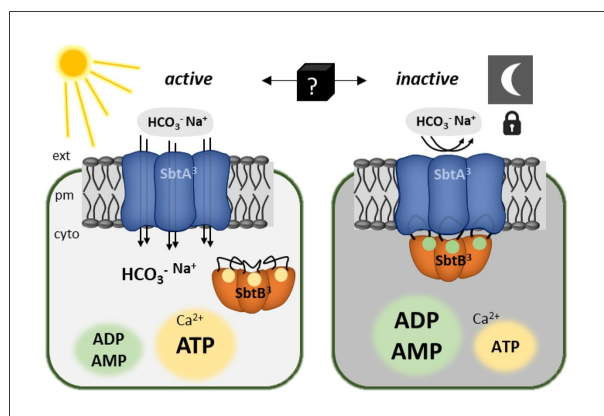


Fig. 8.

Proposed model for regulation of SbtA bicarbonate transport activity involving SbtB by environmental stimuli and the adenylate energy charge (AEC).

In the light, expression of SbtA and SbtB in *Synechococcus elongatus* is induced by low Ci availability. The plasma membrane (pm)-integral SbtA trimer (SbtA₃) actively transports bicarbonate from the external environment (ext) into the cytoplasm (cyto) presumably dissociated from SbtB. SbtA activity depends on Na⁺-symport and requires a cell-inward-directed Na⁺ gradient

with a probable HCO₃⁻:Na⁺ symport stoichiometry of 1:1 through each monomer. Light-stimulated photosynthetic electron transport sustains high levels of intracellular ATP over ADP and AMP, i.e. high AEC, which facilitates binding of Ca²⁺ATP to SbtB trimers (SbtB₃), stabilizing the SbtB T-loops, and SbtB remains dissociated from SbtA. In prolonged darkness, cells operate at a lower AEC. Relatively higher amounts of ADP and AMP replace ATP as ligands in SbtB trimers, which favours open T-loops, and inactive SbtA:SbtB complexes are formed. In such scenario, SbtA reactivation after a transition from dark to light would depend on the rate of increase in AEC and concomitant AEC-dependent dissociation of SbtB.

Conclusion

The adenylate-specific binding of SbtB to SbtA suggests that *in vivo* SbtA trimers become locked into an inactive state when the cellular adenylate energy charge is low, as is the case in low light or darkness. This becomes evident from delay of SbtB-dependent re-activation of SbtA. According to our hypothesis, low AEC-induced binding of SbtB to SbtA provides a curfew mechanism that ensures SbtA activity is tuned to the capacity for photosynthetic C-fixation and tied to the energy resources available. This mechanism may have evolved to override environmental and intrinsic cues that may otherwise induce futile activation/inactivation cycles or activation of SbtA under conditions that demand strategic investment of energy in cell survival rather than maximum growth. Limited light, prolonged nutrient stress or environmental stress may require channelling of resources into maintenance and protective processes instead of maximum photosynthesis and Ci accumulation. It will be of interest to investigate to which extent AEC/SbtB-mediated regulation SbtA has been adopted across cyanobacterial species. The general mechanism of energy charge-sensing via SbtB and its interaction with SbtA is increasingly supported by experimental evidence, however, the precise details are far from being fully understood. Further elucidation of molecular and biochemical mechanisms and further substantiation of the role of the AEC in SbtA:SbtB complex formation *in vivo* under different environmental conditions are challenges for future work. Nonetheless, the link between SbtB as a sensor of the cellular energy status and response regulator of SbtA activity is an important novel insight which may aid development of informed strategies for expression of SbtA-SbtB systems in higher plant chloroplasts.

Materials and Methods

Bacterial and cyanobacterial cell lines

Nucleotide and amino acid sequences were retrieved from the Integrated Microbial Genome database (IMG; <https://img.jgi.doe.gov>) and are referred to by gene ID and Locus_tags. The commercially available *Escherichia coli* strain DH5 α (Invitrogen) was used for cloning and heterologous expression of SbtA and SbtB proteins for binding studies. The HCO $_3^-$ uptake deficient cell line *Synechococcus* Δ CS was derived from wild-type *Synechococcus elongatus* PCC7942 by sequential insertional inactivation of the only two endogenous bicarbonate transporters, BCT1 (*cmpABCD* operon) and SbtA (*sbtAB* operon). BCT1 expression was eliminated by replacement of the *cmpA* (IMG ID: 637799921, Synpcc7942_1488) and part of the *cmpB* (IMG ID: 637799922, Synpcc7942_1489) genes with a selective kanamycin resistance (*kan^R*) cassette (Omata et al., 1990). To this end, a 2.5 kb genomic DNA fragment spanning the *cmpA* and *cmpB* genes was amplified by PCR. The *kan^R* marker was inserted into the central *Bgl*II site of the *cmpAB* fragment, generating gene-specific 3' and 5' flanking sequences for homologous recombination (Table 1) and ligated into pUC18 (pUC18:: Δ *cmpAB*) which is a non-replicating vector in cyanobacteria. The pUC18:: Δ *cmpAB* plasmid was transformed into wild-type and generated the single deletion mutant (Δ C). For the second deletion (Δ S), the full length *sbtA* (IMG ID: 637799907, Synpcc7942_1475) and *sbtB* (IMG ID: 637799908, Synpcc7942_1476) operon was replaced with a chloramphenicol resistance (*cm^R*) marker gene (Dzelzkals et al., 1984) in *Synechococcus* Δ C. Two 1100 and 1400 bp genomic DNA fragments up- and downstream of the *sbtAB* operon, respectively, were amplified by PCR (Table 1), attached as 3' and 5' genomic flanking sequence to the *cm^R* marker gene and inserted into plasmid pUC18 (pUC18:: Δ *sbtAB*), which was then transformed into *Synechococcus* Δ C to generate the double deletion strain (Δ CS). Cyanobacterial transformation was carried out by natural DNA uptake as described previously (Golden and Sherman, 1984). Single cell-derived *kan^R* and/or *cm^R* colonies were selected on the appropriate antibiotic-containing media (kanamycin 100 μ g ml $^{-1}$, chloramphenicol 8 μ g ml $^{-1}$) until Δ C and Δ S deletions had fully segregated, which was verified by PCR using primer pairs Cmp-F/Cmp-R and SbtA-F/SbtB-R, respectively (Table 1).

Table 1.

Primers for generation of the bicarbonate transporter-deficient *Synechococcus* Δ CS mutant.

Genes targeted, inactivation plasmid ^b	Primer ^a	Primer sequence 5' -> 3'	Description
<i>cmpA cmpB</i> , pUC18:: Δ <i>cmpAB</i>	Cmp-F	TTGTCCATCTTTAGAGGCATC	2345 bp genomic fragment spanning <i>cmpA</i> and <i>cmpB</i> . The <i>kan^R</i> cassette was inserted into the central <i>Bgl</i> II site 3' of <i>cmpA</i> .
	Cmp-R	TTCAACAGCAACAAATAAATC	
<i>sbtA sbtB</i> , pUC18:: Δ <i>sbtAB</i>	SbtA-F	TAGGATCCAGACCTTAGCTATTACCAG	1200 bp genomic fragment 3' of <i>sbtA</i>
	SbtA-R	ATGAATTCGATAAAACAGCATGGCAAC	
	SbtB-F	TTGGATCCAGTCCCTCCGCTAAAAG	1375 bp genomic fragment 3' of <i>sbtB</i>
	SbtB-R	AAAAGCTTGGGGGTACTAGAGGTCAAC	

^a Primers used for amplification of genomic DNA from *Synechococcus elongatus* PCC7942.

^b The *kan^R* and *cm^R* marker genes cassettes were inserted into the *cmpAB* genomic fragment and the *sbtA* and *sbtB* flanking regions, respectively, and cloned into the pUC18 vector.

For protein expression, *sbtA* and *sbtB* genes from *Synechococcus elongatus* PCC7942 and *Cyanobium* sp. PCC7001 (IMG ID: 647590134, CPCC7001_1784 and IMG ID: 647590133, CPCC7001_1671, respectively) were synthesized (GenScript, New Jersey, USA) and inserted into the *SacI/XbaI* restriction sites of plasmids pSE2-1 and pSE4-1 which both carry a spectinomycin resistance (*sp^R*) marker gene (Price et al., 2004; Du et al., 2014) and contain origins of replications for maintenance in *E. coli* and in *Synechococcus elongatus*. Genes of interest were expressed from pSE2-1 under control of the isopropyl β -D-1-thiogalactopyranoside (IPTG)-inducible *lacZ* promoter and *lacIQ* repressor and from pSE4-1 using their native *sbtA* and *sbtB* promoters. The SbtB proteins were C-terminally tagged with a HA epitope (YPYDVPDYA) for immunodetection and 5 or 6 His residues for affinity purification in IMAC assays. For expression of SbtA alone, the appropriate *sbtB* genes were removed by restriction digestion and re-ligation from the *sbtAB* gene inserts in pSE2-1 and pSE4-1. Transformed *E. coli* and cyanobacterial cells were selected on solid media containing 100 and 10 $\mu\text{g ml}^{-1}$ spectinomycin, respectively. Cell lines and expression vectors used in this study are listed in Table 2.

Table 2.

E. coli and cyanobacterial cell lines and plasmids used for protein expression and physiological analyses of the SbtA and SbtB proteins from *Cyanobium* sp. PCC7001 and *Synechococcus elongatus* PCC7942.

Organism	Plasmid::Gene(s) of interest ^a	Promoter(s)	Proteins expressed
<i>E. coli</i> DH5 α	pSE2-1::sbtAsbtBHAH6-7942	<i>lacZ</i>	SbtA ₇₉₄₂ SbtB ₇₉₄₂ -HAHis
<i>E. coli</i> DH5 α	pSE2-1::sbtAsbtBHAH5-7001	<i>lacZ</i>	SbtA ₇₀₀₁ SbtB ₇₀₀₁ -HAHis
<i>Synechococcus</i> Δ CS	pSE4-1::sbtA-7942	native <i>sbtAB</i> -7942 ^b	SbtA ₇₉₄₂
<i>Synechococcus</i> Δ CS	pSE4-1::sbtAsbtB-7942	native <i>sbtAB</i> -7942 ^b	SbtA ₇₉₄₂ SbtB ₇₉₄₂
<i>Synechococcus</i> Δ CS	pSE4-1::sbtAsbtBHAH6-7942	native <i>sbtAB</i> -7942 ^b	SbtA ₇₉₄₂ SbtB ₇₉₄₂ -HAHis
<i>Synechococcus</i> Δ CS	pSE4-1::sbtAsbtBHAH6-7001	native <i>sbtAB</i> -7001 ^c	SbtA ₇₀₀₁ SbtB ₇₀₀₁ -HAHis

^a pSE2-1 and pSE4-1 contain a spectinomycin resistance cassette (*sp^R*) for selection.

^b *sbtA*-7942 and *sbtB*-7942 are transcribed in opposite directions away from each other. The intergenic region between the ATG-translation start codons contains used for expression contains the promoters both genes.

^c *sbtA*-7001 and *sbtB*-7001 are transcribed in the same direction in tandem. The promoter is contained within the 500 bp upstream of the GTG-translation start for *sbtA*.

Growth of *E. coli* and cyanobacteria

For maintenance, mutant selection and protein expression, *E. coli* DH5 α derived cell lines were grown at 37 °C in Lysogeny broth (LB) supplemented with 100 $\mu\text{g ml}^{-1}$ spectinomycin as described previously (Du et al., 2014), and expression of SbtA and SbtB proteins was induced for 3 h with 1 mM IPTG. *Synechococcus* Δ CS cell lines were grown at 30 °C in modified BG-11 medium (buffered with 20 mM HEPES-KOH pH 8.0 (Maeda et al., 2002; Woodger et al., 2003) in high Ci (2-4% (v/v) CO₂ enriched air) or low Ci (ambient air, 400 ppm CO₂) under constant light (70 $\mu\text{mol m}^{-2} \text{s}^{-1}$). Antibiotics were supplied as appropriate (100 $\mu\text{g ml}^{-1}$ kanamycin, 8 $\mu\text{g ml}^{-1}$ chloramphenicol, 100 $\mu\text{g ml}^{-1}$ spectinomycin). Protein expression of SbtA and SbtB was induced by exposing high Ci-grown cyanobacterial cultures for 4 h to low Ci. For growth

assays, dilution series of low Ci-induced cultures grown to mid-log phase were adjusted to OD₇₃₀ of 0.2 and a dilution series spotted onto agar plates containing growth medium with antibiotic supplements. Plates were then incubated in air under 50 $\mu\text{mol m}^{-2} \text{s}^{-1}$ white light (photoperiod, 12 h light: 12 h dark) for 10 days. Final biomass accumulation was recorded as scanned images.

Extraction of native proteins

The method for the isolation of membrane-enriched fractions of native proteins from *E. coli* and *Synechococcus* ΔCS was adapted from previous work (Du et al., 2014). After induction of protein expression, cultures of *E. coli* or *Synechococcus* ΔCS were harvested in mid-log phase by centrifugation (5 min, 6000 *g*, 23 °C). Cell pellets were washed once with HEPES lysis buffer (50 mM HEPES, 100 mM NaCl, 5% (v/v) glycerol) and treated with 300 U/ μl r-Lysozyme (Sigma, USA) for 15 min at 23 °C. Cells were broken with 0.1 mm Zirconia-silica beads (Biospec Products) in a mini-beadbeater 16 (Biospec Products) at 3450 oscillations min^{-1} for 6 min in HEPES IMAC binding buffer (50 mM HEPES, 100 mM NaCl, 25 mM imidazole, pH 8.0) and 0.1% (v/v) protease inhibitor cocktail (Sigma, USA) and centrifuged for 30 s at 16000 *g* to precipitate cell debris and beads at room temperature. The supernatant was then centrifuged (15 min, 16000 *g*, 4 °C) to pellet the membrane-enriched fraction. Membrane proteins were solubilized in IMAC HEPES binding buffer (50 mM HEPES, 100 mM NaCl, 25 M imidazole, pH 8.0) supplemented with 1% (w/v) *n*-dodecyl β -D-maltoside (DDM) by gentle agitation on a rotator for 30 min at 23 °C, and insoluble matter precipitated by centrifugation (5 min, 16000 *g*, 4°C) prior to further analysis.

Immobilized metal affinity chromatography (IMAC) protein binding assays

The association of SbtA and HAHis-tagged SbtB proteins was analysed in IMAC binding assays adapted from previous work (Du et al., 2014). Native protein extracts from *E. coli* or *Synechococcus* ΔCS were incubated in IMAC HEPES binding buffer supplemented with potential effector molecules and Ni-charged iminodiacetic acid Profinity™ IMAC Resin (Bio-Rad, USA) on a low speed rotator for 1 h at 23°C. This allowed the His-tag on SbtB to bind to the resin and SbtA to associate with the immobilized SbtB protein. The resin had been prewashed three times with IMAC HEPES binding buffer, and one volume of resin was added to four volumes of the reaction mixture. Depending on the experiment, varying concentrations of MgCl_2 , CaCl_2 , NaHCO_3 , Na_2HPO_4 , RUBP, 3PGA, 2OG, ATP, ADP, AMP and cAMP were added to the protein extract-resin mixture as potential effector molecules. Standard binding assays were performed with 200 nM CaCl_2 unless stated otherwise. Further details on effector molecules and adenylnucleotide ratios are provided in the figures and captions. The reaction mixture was transferred onto columns, washed with five volumes IMAC HEPES binding buffer by gravity flow through, which removed unbound protein and effector molecules. SbtB was then eluted from the resin with IMAC HEPES elution buffer (50 mM HEPES, 300 mM NaCl, 250 mM imidazole, pH 8.0). The eluted fraction contained the proportion of SbtB protein immobilized on the beads and the amount of SbtA protein retained due to binding to SbtB. IMAC purified SbtA:SbtB complexes were subjected to immunoblot analysis.

Immunodetection of SbtA and SbtB proteins and quantification of SbtA:SbtB binding

Aliquots of membrane-enriched and eluted IMAC fractions were supplemented with 2.5% (v/v) β -mercaptoethanol and sample loading buffer (62.5 mM Tris pH 8.0, 2% SDS, 10%

glycerol, 0.02% (w/v) Coomassie G250, 0.0025% (w/v) Phenol red), heated for 3 min at 80 °C, centrifuged (30 s, 12000 g, 23°C) and separated on linear gradient 4-12% Bis-Tris polyacrylamide gels (NuPAGE® Novex®, Invitrogen, USA) in MES running buffer (50 mM MES, 50 mM Tris base, 0.2% SDS, 1 mM EDTA free acid). Proteins were transferred onto 0.45 µm PVDF (polyvinylidene fluoride) membrane using a semi-dry blotter (Bio-Rad, USA) and Tris-Glycine transfer buffer (48 mM Tris base, 38 mM glycine, 0.038% SDS, 20% methanol). Immunological detection was performed in Tris-Saline buffer (20 mM Tris, 137 mM NaCl, 0.1% Tween 20, pH 7.5) with 5% (w/v) skim milk powder as blocking reagent. SbtA protein was detected with a polyclonal antibody generated against a conserved 14 amino acid-epitope (Agrisera, Sweden). The HAHis-tagged SbtB proteins were detected with monoclonal antibodies against the HA-epitope (Sigma, USA). Secondary, alkaline phosphatase-conjugated antibodies (Bio-Rad, USA) were used for fluorometric detection of the immune-reactive bands with the AttoPhos system (Promega, USA). The fluorescence signal was visualized with a ChemiDoc™ MP imaging system (Biorad, USA). The relative signal intensities were quantified based on global background-corrected pixel volumes using ImageLab software (Bio-Rad, USA). Amounts of SbtA protein retained in complexes by binding to SbtB (pixel volumes; y) were plotted against adenyl nucleotide ratios (AR; x), using OriginPro v2020 software (OriginLab Corp., USA). The non-linear correlation was fitted by a type 1 sigmoidal-logistic function ($F(x) = y = A/(1 + e^{-k*(x-x_c)})$), with A as maximum amount of SbtA bound, k as rate (binding) coefficient, and x_c as the AR at the inflection point. The adenyl nucleotide ratios corresponding to 50% (AR_{50}) and 10% (AR_{10}) of SbtA bound to SbtB were estimated from the sigmoidal logistic fitted curves normalized to residual SbtA detected at $AR=1$.

Photosynthetic analysis by membrane inlet mass spectrometry (MIMS) measurements

Net photosynthetic O_2 evolution and Ci uptake rates were calculated from changes in the concentrations of dissolved CO_2 and O_2 in aqueous cyanobacterial cell suspensions measured in a closed custom-built, temperature-controlled cuvette attached to a mass spectrometer (Isoprime IRMS series, Micromass, UK) as described previously (Badger and Andrews, 1982; Badger et al., 1985; Price et al., 2004). The inside of cuvette was separated from the inlet to the mass spectrometer by a gas permeable membrane. Low Ci -induced cell cultures were harvested by centrifugation (4 min, 4000 g, 23 °C), washed once with CO_2 -free BTP (1,3-bis[tris(hydroxymethyl)methylamino]propane) assay buffer (20 mM BTP buffered modified Bg-11(- NO_3), 20 mM NaCl, pH 8.0), and resuspended at cell densities equal to 4 µg ml^{-1} Chl a. Chl a content was determined spectrophotometrically in 90% (v/v) methanol extracts (Porra et al., 1989). Measurements were performed at 30 °C which is equivalent to the growth temperature for cell cultures. During MIMS measurements, CO_2 uptake was inhibited with 500 µM ethoxymethylamine (EZ). Steady state photosynthetic O_2 evolution in response to light intensity was recorded for cell suspension cultures equilibrated with 1 mM HCO_3^- which constitutes > 98% of the available Ci at pH 8. Illumination (KL1500 HAL light source, Schott, Germany) was increased stepwise from 0 to 15, 50, 200, 490, 850 and 1100 µmol photons $m^{-2} s^{-1}$. Respiration rates were estimated from O_2 consumption in the dark. Prior to MIMS measurements of Ci response curves, cell cultures were depleted for Ci by bubbling with N_2 in white light (400 µmol $m^{-2} s^{-1}$) for at least 10 min and photosynthetic O_2 evolution rates recorded following incremental additions of bicarbonate. Steady state O_2 evolution rates (V) normalized to Chl a contents were plotted against Ci substrate (S) concentrations, and resulting curves were fitted by the Michaelis-Menten equation, $V = (V_{max} [S])/(k_{0.5} + [S])$, using OriginPro v2021 software. Maximum O_2 evolution rates (V_{maxO_2}) were calculated as an approximation for maximum bicarbonate uptake by SbtA. Affinity of SbtA for HCO_3^- was deduced from the Ci concentration supporting half-maximal activity ($k_{0.5Ci}$). Effects of 300 s-illuminations with high (HL, 1400 µmol $m^{-2} s^{-1}$) and low (LL, 200

$\mu\text{mol m}^{-2} \text{s}^{-1}$) white light on HCO_3^- uptake and steady state O_2 evolution rates were determined in cell cultures that had been equilibrated in the dark after addition of aqueous NaHCO_3 solution to a final concentration of 250 or 500 μM .

Ci uptake measurement by silicon oil centrifugation-filtration

Light- and dark stimulated uptake of active species of $^{14}\text{C}[\text{HCO}_3^-]$ in cyanobacterial cell cultures was measured using the silicon oil centrifugation-filtration method as described previously (Kaplan et al., 1980; Price and Badger, 1989). Low Ci-induced cell cultures were prepared as described for MIMS analyses. Cells were pelleted by centrifugation (4 min, 4000 g, 23 °C) and resuspended in CO_2 -free BTP at pH 8 or pH 9 at densities equal to 4 $\mu\text{g ml}^{-1}$ Chla. Ci uptake was determined in 200 μl aliquots of cells supplemented with 500 μM EZ that were transferred into translucent centrifuge tubes on top of a silicon oil layer (mixture of AR200 : AR20 3.5 : 4 (v/v)) which separated the cells from the kill solution (50% methanol, 2 N NaOH) in the bottom of the tube. For low light (30 $\mu\text{mol photons m}^{-2} \text{s}^{-1}$) activation experiments, cells were equilibrated with 500 μM $^{14}\text{C}[\text{HCO}_3^-]$ substrate mix (aqueous NaHCO_3 solution spiked with radioactive $^{14}\text{C}[\text{NaHCO}_3]$, pH 9.5) for 60 s either in the dark (dark control) or after exposure to low light for 30 or 60 s following a transfer from the dark. Uptake was terminated by rapid filtering of the cells into the kill solution during 20 s centrifugation at set maximum speed (microfuge B, Beckman, USA). For dark-inactivation experiments, cell cultures were depleted of Ci in the light (400 $\mu\text{mol m}^{-2} \text{s}^{-1}$) prior to initiation of HCO_3^- uptake by adding 500 μM $^{14}\text{C}[\text{HCO}_3^-]$ substrate mix, either in the light (400 $\mu\text{mol m}^{-2} \text{s}^{-1}$), or at various time points in the dark after cells had been transferred from the light into the dark for 15, 30, 45, 60, 90 and 180 s. Cells were incubated with ^{14}C -labelled substrate for 30 s, and uptake was terminated by 20 s centrifugation. Tubes were immediately frozen in dry ice. The tips containing the cell pellets were cut off, resuspended in 2 N NaOH and mixed with 5 volumes scintillation fluid (Ultima Gold™ XR, PerkinElmer) for ^{14}C -CPM counting in a TRI-CARB liquid scintillation counter (PerkinElmer, USA). Ci uptake rates were calculated based on the specific activity of the $^{14}\text{C}[\text{HCO}_3^-]$ substrate mix. To test the inhibitory effect of EZ, cells were incubated with varying concentrations of EZ (0, 250, 375, 500, 750, 1000 μM) for 5 min in the light before adding 500 μM $^{14}\text{C}[\text{HCO}_3^-]$ substrate mix for 30 s-uptake measurements.

Bioinformatics

Homologous SbtA and SbtB proteins were identified in BLASTP (Altschul et al., 1990) searches and multiple pairwise amino acid sequence alignments generated in Geneious Prime 2021 (<https://www.geneious.com>) with MUSCLE software (Edgar, 2004). Further details on aligned proteins is given in **Supporting Information Tables S1 and S2**. The three-dimensional domain homology models of the SbtB₇₉₄₂, SbtA₇₉₄₂ and SbtA₇₀₀₁ monomers were generated using AlphaFold2 (ColabFold v1.5.2: AlphaFold2 using MMseqs2) (Mirdita et al., 2022). Three-dimensional structure visualization and comparisons were performed with UCSF Chimera v.1.16 (<http://www.cgl.ucsf.edu/chimera>) (Meng et al., 2006) and UCSF ChimeraX (<https://www.rbvi.ucsf.edu/chimerax>) (Pettersen et al., 2021). The overall folds of SbtB₇₉₄₂ and SbtB₇₀₀₁ chain A with ADP (PDB: 6MMC) or Ca^{2+} and ATP ligands (PDB: 6MM2) proteins were compared by superposition onto each other and the cryo-EM structure of SbtB₆₈₀₃ (PDB: 7EGK, chains B, D, or F). The SbtA₇₉₄₂ and SbtA₇₀₀₁ homology models were compared by superposition onto each other and the cryo-EM structure of SbtA₆₈₀₃ (PDB: 7EGK, chains A, C or E).

Statistics

Statistically significant differences between data sets were determined by performing a one-way analysis of variance (ANOVA) and post-hoc Tukey's Honest Significant Difference (HSD) test (Tukey, 1949), using a type I error level $\alpha=0.05$ ($P=0.05$) as threshold.

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Author Contributions

BF, BM and GDP: Conceptualization and Methodology. BF, BM and LR: Investigation. BF: Formal data analysis. BF: Writing - original manuscript draft. All authors: Writing-review & editing. GDP: Funding acquisition.

Conflict of Interest

No conflicts of interest are declared.

Data availability

All data supporting this study are available within the main manuscript and the supplementary information published online. Any data not shown are available from the corresponding author upon request.

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Reviewer #1 (Public Review):

Using an immobilised metal affinity chromatography (IMAC)-based assay coupled with Western blot immunodetection analysis, SbtB, the regulatory protein for SbtA activity, is shown in itself to be regulated by the local adenylate energy charge (AEC), with inhibitory binding of SbtB to SbtA disfavoured at high ATP:ADP ratios. Such conditions are expected to be encountered during steady-state photosynthesis with the associated cellular demand for Ci and SbtA activity.

By homology with ATP-binding PII proteins, ATP is proposed to interact with a loop region of SbtB, changing its conformation on binding and inhibiting the formation of the (inactive) SbtA:SbtB complex. On the basis of this, the authors propose that SbtB acts an AEC-sensing

'curfew' protein for SbtA activity, tuning bicarbonate import by this protein for situations when carbon fixation would be physiologically (and energetically) advantageous. As SbtA is a $\text{HCO}_3^-/\text{Na}^+$ symporter, Na^+ homeostasis would also be controlled by regulation of this transporter.

The IMAC assay used to monitor SbtA:SbtB complex stability as a function of AEC seems robust, is relatively straightforward and may be of interest to other researchers investigating adenylate-sensing protein reaction partners (with the usual caveats on extrapolating in vitro results to living systems, as noted by the authors).

In this study, SbtA regulation was also investigated in vivo in a *Synechococcus* HCO_3^- -transporter knockout mutant via measurement of labelled HCO_3^- uptake and overall photosynthetic performance (MIMS-monitored O_2 evolution as a function of PAR). Here, SbtB was inferred to regulate SbtA activity during the induction of photosynthesis (i.e. at low ATP:ADP) and not when photosynthesis was fully activated and in a steady-state condition. SbtA inactivation on a light-dark transition was also demonstrated in vivo irrespective of the presence SbtB, indicative of additional regulatory pathways affecting the activity of this transporter. These conclusions seem to be well-supported by the presented data.

Reviewer #2 (Public Review):

Inorganic carbon (Ci) uptake by autotrophic organisms is often the rate-limiting process in overall photosynthetic productivity. Aquatic autotrophs including the cyanobacteria have evolved elaborate and metabolically expensive, yet very efficient CO_2 concentrating mechanisms (CCMs) to over-come this limitation. The work examines the regulation of SbtA, which is a high affinity sodium dependent symporter. Current evidence suggests that this SbtA is highly regulated both at the transcriptional and post-transcriptional levels. For example, the *sbtA* gene is transcriptionally upregulated under conditions of inorganic carbon limitation and the transport activity of the expressed SbtA protein is apparently regulated allosterically by multiple factors, including those exerted by the binding of the small trimeric protein, SbtB. SbtB is a PII-type regulator that conditionally binds to the cytoplasmic face of the trimeric SbtA to form a hetero-complex apparently inactivating SbtA to which it is bound. The factors affecting this interaction remains to be clarified, but it is already clear that there is considerable complexity that needs to be unraveled since as with other PII proteins, multiple effector molecules act as ligands.

Using a novel protein-protein interaction assay combined with physiological analysis of various mutants, the authors present new information on the regulation of SbtA from *Cyanobium* sp. PCC7001 and *Synechococcus elongatus* PCC7942. Because of their novelty, additional validation may be important to establish their validity, yet they do appear to be robust overall. The work builds on earlier studies indicating negative regulation of SbtA and helps clarify other work, including detailed analysis of the orthologous, albeit somewhat more complex protein from *Synechocystis* PCC6803. The key significance of the present findings is that the energy charge of the adenylate system, a ubiquitous metabolic control mechanism in the biological world, is the prime and perhaps overriding regulatory parameter governing of SbtA activity. Based on this a model for the diurnal control transporter activity was proposed based on energy charge.

Reviewer #3 (Public Review):

The regulation of transporters in many physiological systems is poorly known. Here, Forster and colleagues describe how activity of an inorganic carbon transporter, SbtA, in the

bacterial carbon concentrating mechanism is regulated by the PII protein SbtB. Although there is now significant structural knowledge of the system and many potential SbtB-regulating small molecule effectors are known, Förster and colleagues clarify, how the adenylate charge in the cell, rather than any single metabolite, is the important regulatory effector. This is critical for the endogenous function, as the cyanobacterial host undergoes dramatic changes in adenylate charge over the course of a diurnal cycle and this result explains how the channel is regulated to efficiently function in CO₂ assimilation. The manuscript is generally clear and the data generally supportive of the conclusions as written. However, there are several instances where additional clarification and/or experiments are needed to confirm the major findings of the paper.